

WEEK-1 PROJECT REPORT

TELECOM DATA GOVERNANCE & SECURITY

Internship Domain:

Telecom Data Governance & Security

Duration:

6 Weeks

Submitted by:

DUDATI RASIKA

Week:

Week1-Telecom Data Governance Strategy and Planning

Introduction

The telecommunications industry is one of the most data-intensive sectors. Telecom organizations continuously generate and process information through customer registrations, mobile subscriptions, billing systems, call and internet usage, network operations, customer support, mobile applications and connected devices.

The growth of 4G, 5G, cloud computing, IoT and analytics has increased the volume and complexity of telecom data. While this data provides significant business value, it also creates challenges related to data accuracy, consistency, security, privacy, access control, compliance and lifecycle management.

Data governance provides a structured approach for managing these challenges. It defines policies, responsibilities, standards and processes that determine how data should be collected, stored, processed, accessed, shared, protected, retained and deleted.

This project proposes a **Telecom Data Governance & Security** framework for a hypothetical organization, ABC Telecom. The Week 1 task focuses on developing the strategic foundation for the governance program, including objectives, scope, stakeholders, resources, risks, implementation planning and performance measurement.

The official internship task describes Week 1 as developing a comprehensive telecom data governance strategy and planning document that can serve as a blueprint for telecom companies.

2. Project Overview

2.1 Project Title

Telecom Data Governance & Security

2.2 Week 1 Task

Telecom Data Governance Strategy and Planning

2.3 Project Purpose

The purpose of this project is to develop a structured framework for managing telecom data in a secure, consistent and accountable manner.

The framework focuses on:

- Data quality
- Data security
- Privacy
- Regulatory compliance
- Data ownership
- Access management
- Data lifecycle management
- Risk management
- Monitoring and evaluation

2.4 Hypothetical Organization

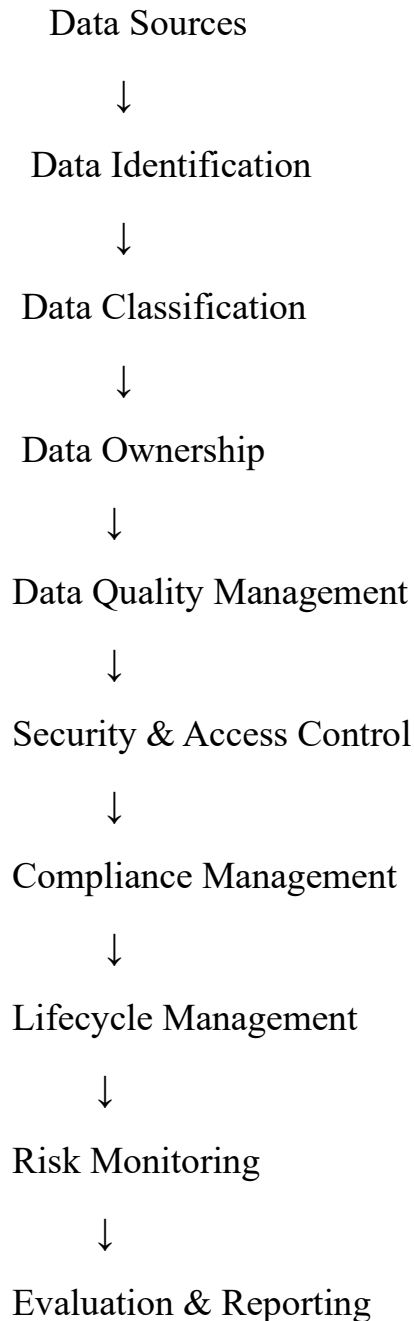
For this project, **ABC Telecom** is used as a hypothetical telecom organization.

ABC Telecom is assumed to provide:

- Mobile services
- 4G and 5G connectivity
- Broadband services
- Enterprise services
- Mobile applications

2.5 Project Concept

The project proposes that ABC Telecom establish a centralized governance approach in which:



2.6 Practical Extension

Although Week 1 is primarily a strategy and planning task, the proposed framework can later be implemented through a web-based **Telecom Data Governance & Security Portal**.

Possible modules include:

- Dashboard
- Data Catalogue
- Data Classification
- Data Quality Monitoring
- User and Role Management
- Access Requests
- Audit Logs
- Compliance Monitoring
- Reports

3. Telecom Data Landscape and Background Research

3.1 Telecom Data Environment

Telecom companies generate information from multiple sources.

Customer Data

Includes customer identification and contact information.

Subscriber Data

Includes mobile numbers, SIM information and subscription details.

Billing Data

Includes plans, invoices, payments and billing status.

Network Data

Includes network performance, traffic and infrastructure information.

Usage Data

Includes voice, SMS and internet usage information.

Device Data

Includes device identifiers and device-related information.

Analytics Data

Includes reports, dashboards and analytical datasets.

3.2 Major Telecom Data Sources

System	Data Generated
CRM	Customer information
Billing	Payments and invoices
Network Systems	Network performance
Mobile Application	User activity
Customer Support	Complaints and service requests
Data Warehouse	Consolidated information
Analytics Platform	Reports and insights
Vendor Systems	External service information

3.3 Current Data Governance Trends

5G Data Management

5G networks create large volumes of network and usage information. Governance is required to manage this information consistently.

IoT

Connected devices continuously generate information, increasing the need for data classification, monitoring and lifecycle controls.

Cloud Data Governance

Cloud environments require appropriate controls for data access, security, monitoring and management.

Artificial Intelligence and Analytics

AI systems depend on reliable data. Poor-quality information can negatively affect analytical results.

Data Catalogues

Data catalogues provide visibility into available datasets, their owners, definitions and locations.

Metadata Management

Metadata helps organizations understand the meaning, source and characteristics of data.

Data Lineage

Data lineage helps trace data from its source through transformation and use.

Automated Data Quality

Automation can continuously identify data-quality problems such as duplicates, missing values and invalid records.

4. Problem Statement and Need for Data Governance

4.1 Problem Statement

ABC Telecom is assumed to have data distributed across multiple systems. The absence of a coordinated governance framework can result in inconsistent data, unclear ownership, security risks, compliance difficulties and inefficient data management.

4.2 Major Problems

Inconsistent Data

Customer information may differ between CRM and billing systems.

Poor Data Quality

Data may contain duplicate, missing, invalid or outdated information.

Unclear Ownership

Departments may not know who is responsible for particular datasets.

Unauthorized Access

Users may have unnecessary access to sensitive information if permissions are not properly managed.

Compliance Risk

Failure to maintain appropriate processes can create compliance gaps.

Data Silos

Information may remain isolated in different systems, making it difficult to obtain a consistent organizational view.

4.3 Need for Governance

A formal governance framework is required to:

- Establish accountability
- Improve data quality
- Protect sensitive information
- Support compliance
- Standardize data processes
- Reduce risks
- Improve decision-making

5. Strategic Objectives and Scope

5.1 Strategic Objectives

Objective 1: Improve Data Quality

Ensure data is accurate, complete, consistent, valid, unique and timely.

Objective 2: Strengthen Data Security

Protect sensitive data through appropriate authentication, authorization, access controls, encryption and monitoring.

Objective 3: Support Compliance

Establish processes that support applicable regulatory, privacy and organizational requirements.

Objective 4: Establish Data Ownership

Assign clear responsibility for critical datasets.

Objective 5: Improve Data Management

Standardize data collection, storage, processing, usage, sharing, retention and deletion.

Objective 6: Improve Data Discoverability

Maintain information about datasets, owners, definitions and sources.

Objective 7: Enable Business Value

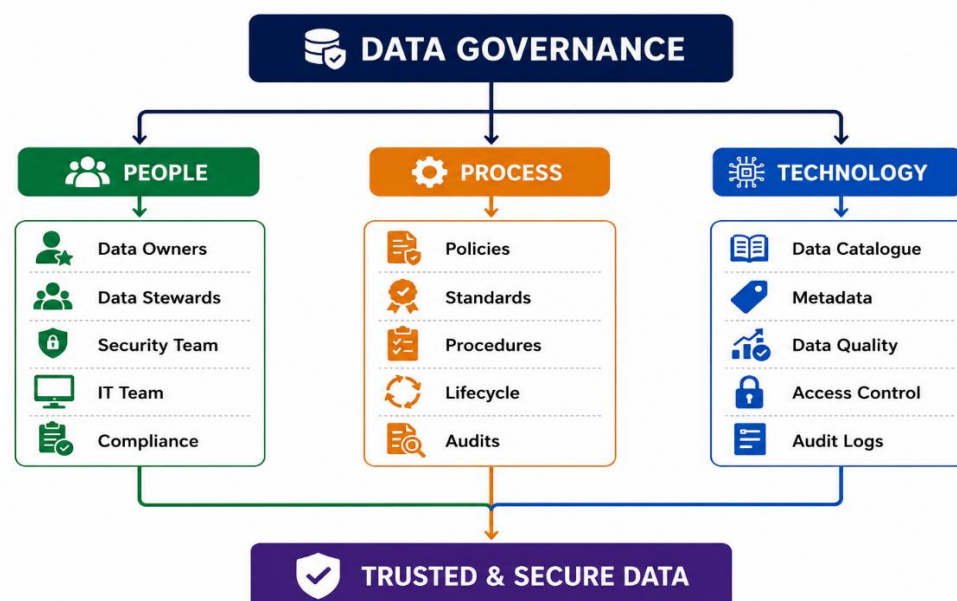
Provide trusted data for reporting, analytics and decision-making.

5.2 Project Scope

The framework covers:

- Customer data
- Subscriber data
- Billing data
- Network data
- Usage data
- Device data
- Employee data
- Vendor data
- Product data
- Analytics data

6. Proposed Data Governance Framework



6.1 People

People establish accountability.

Important roles include:

- Executive Sponsor
- Data Governance Council
- Data Governance Office
- Data Owners
- Data Stewards
- IT/Data Custodians
- Security Team
- Compliance Team

6.2 Process

Processes establish:

- Data policies
- Data standards
- Data-quality procedures
- Access procedures
- Security procedures
- Retention procedures
- Audit procedures
- Incident procedures

6.3 Technology

Technology supports:

- Data catalogue
- Metadata
- Data quality

- Access control
- Audit logging
- Monitoring
- Reporting

7. Stakeholder Analysis and Responsibilities

Stakeholder	Responsibility	Interest
Senior Management	Strategic direction	High
Governance Council	Governance decisions	High
Governance Office	Program coordination	High
Data Owners	Dataset accountability	High
Data Stewards	Data quality	High
IT Team	Technical infrastructure	High
Security Team	Security controls	High
Compliance Team	Compliance	High
Customer Service	Customer information	High
Finance	Billing information	Medium
Marketing	Analytics	Medium
Vendors	External services	Medium

Responsibility Structure

Executive Management



Data Governance Council



Data Governance Office



Data Owners



Data Stewards



IT / Security / Compliance

8. Data Quality, Security and Compliance Strategy

8.1 Data Quality

The framework uses six dimensions:

Dimension	Dimension
Accuracy	Correctness of information
Completeness	Required information is available
Consistency	Information is consistent across systems
Timeliness	Information is sufficiently current
Validity	Information follows defined rules
Uniqueness	Duplicate records are minimized

Data Quality Process

Identify



Define Rules



Validate



Detect Issues



Correct



Revalidate



Monitor

8.2 Security

The proposed security strategy includes:

Authentication

Users must verify their identity before accessing the system.

Authorization

Access is granted according to approved permissions.

Role-Based Access Control

Permissions are assigned according to roles.

Least Privilege

Users receive only the access necessary for their responsibilities.

Encryption

Sensitive information should be appropriately protected during storage and transmission.

Data Masking

Sensitive information can be partially hidden.

Example:

9876543210 → 98765XXXXX

Audit Logging

Important activities should be recorded.

8.3 Compliance

The compliance strategy includes:

1. Identify applicable requirements.
2. Map requirements to policies.
3. Assign responsibility.
4. Monitor compliance.
5. Maintain evidence.
6. Conduct periodic reviews.
7. Identify gaps.
8. Implement corrective actions.

9. Implementation Strategy and Timeline

A proposed implementation can be divided into six phases.

Phase	Timeline	Activities
Assessment	Months 1–2	Identify systems, datasets and gaps
Strategy	Months 3–4	Develop policies and governance structure
Pilot	Months 5–6	Implement selected governance controls
Expansion	Months 7–9	Extend governance to additional domains
Optimization	Months 10–11	Improve and automate processes
Evaluation	Month 12	Measure results and plan improvements

10. Resource Allocation and Risk Management

10.1 Human Resources

The proposed program may require:

- Governance Manager
- Data Owners
- Data Stewards
- Data Analysts
- IT Specialists
- Security Specialists
- Compliance Specialists
- Project Manager

10.2 Technology Resources

Possible technology requirements include:

- Database systems

- Data catalogue
- Data-quality tools
- Access-management systems
- Monitoring tools
- Audit systems
- Reporting systems

10.3 Training

Employees should receive training on:

- Data governance
- Data quality
- Security
- Privacy
- Data handling
- Incident reporting

10.4 Risk Management

Risk	Likelihood	Impact	Mitigation
Poor data quality	High	High	Validation and monitoring
Unauthorized access	Medium	High	RBAC and access reviews
Data leakage	Medium	High	Encryption and masking
Unclear ownership	High	High	Assign Data Owners
Compliance failure	Medium	High	Periodic compliance reviews
Legacy systems	High	High	Phased implementation
Employee resistance	Medium	Medium	Training
Data duplication	High	Medium	Quality controls
Integration issues	Medium	High	Testing and standards

11. Performance Metrics, Expected Outcomes and Assumptions

11.1 KPIs

The following are illustrative targets:

KPI	Target
Critical datasets with assigned owners	100%
Critical datasets classified	100%
Data-quality score	≥95%
Quality issues resolved within SLA	≥90%
Policy compliance	≥95%
Access reviews completed	100%
Governance incidents	Reduce over time
Employee governance training	≥95%

11.2 Expected Outcomes

The framework is expected to provide:

- Improved data quality
- Better data consistency
- Clear data ownership
- Stronger security
- Better compliance
- Reduced risks
- Improved reporting
- Better decision-making
- Improved lifecycle management
- Greater accountability
- Greater trust in organizational data

11.3 Assumptions

1. ABC Telecom is a hypothetical organization.
2. The organization operates multiple systems containing telecom data.
3. The organization processes information requiring appropriate protection.
4. Management supports the governance program.
5. Data Owners and Data Stewards can be assigned.
6. Sufficient technical and human resources are available.
7. The implementation roadmap is illustrative.
8. Actual legal and regulatory requirements must be validated before implementation.
9. The proposed web portal is a future technical implementation.

12. Recommendations

Based on the proposed framework, the following recommendations are made:

1. Establish a formal Data Governance Council.
2. Assign Data Owners and Data Stewards.
3. Create organization-wide data-quality standards.
4. Implement data classification.
5. Introduce role-based access control.
6. Conduct regular access reviews.
7. Establish a centralized data catalogue.
8. Implement continuous data-quality monitoring.
9. Maintain audit logs.
10. Establish data-retention and secure-deletion policies.
11. Conduct periodic governance audits.
12. Provide employee training.
13. Monitor governance KPIs.

13. Conclusion

The **Telecom Data Governance & Security** project provides a structured approach for managing data within a telecommunications organization. The proposed framework addresses data quality, security, compliance, ownership, accountability, lifecycle management, risk management and performance measurement.

The project demonstrates that effective data governance is broader than cybersecurity alone. Security is an important component, but governance also requires clearly defined people, processes, policies, standards and supporting technology.

The proposed People–Process–Technology framework provides a foundation for establishing reliable, secure and accountable data-management practices. It can help telecom organizations improve data quality, reduce risks, strengthen security, support compliance and increase trust in organizational information.

The framework can subsequently be translated into a practical web-based Telecom Data Governance & Security Portal containing features such as data cataloguing, classification, quality monitoring, access management, audit logging, compliance monitoring and reporting.

Therefore, the proposed strategy serves as a foundation for both **strategic governance planning and future practical implementation**.

14. References

For the final Word document, this section should contain the **actual publicly available sources you used for your research**.

Use a consistent format such as **IEEE** or **APA**.

Recommended source categories:

- Telecom industry standards
- Data governance frameworks
- Data-quality standards
- Cybersecurity frameworks
- Privacy and data-protection regulations
- Telecom regulatory guidance
- 5G and IoT governance research
- Cloud data governance resources
- Data lifecycle management standards